FEVER against LOFT attacks: summary of observations from the final testbed campaign

14 November 2026

1. What was evaluated

FEVER combines a detector trained by lossless (histogram-based) federated XGBoost across three SDN controllers with an eviction and ingress-blocking response (EvictBlock). The controllers C1, C2 and C3 each trained on their own traffic trace and their own two attacks; no rows or labels left a controller. FEVER was compared with six published defenses, reimplemented from their papers: FTMaster, FTOP, FTODefender, FloRa, DMS-BTT and rDefender.

The campaign ran on a Mininet and Ryu testbed with a three-switch tree, where the core switch s11 overflows. Two test traces were used, T1B and UHt (a trace no controller trained on), and seven attacks: G2, Spoof, Pulse, Mimic, Aware, Slow and EVX. It comprised 468 runs of 300 s with 3 to 5 seeds per cell, plus 64 reruns of DMS-BTT after a correction (amendment 3). Hypotheses H1 to H7 were preregistered and tested with one-sided paired sign tests, Holm-corrected over 33 tests.

Every attack follows the low-rate flow-table overflow pattern of DMS-BTT: in each burst the attacker re-sends packets on all its live flows, so that they are refreshed before the switch's 10 s idle timeout, and opens S new flows, until it holds M flows. Bursts are sent at 1,500 packets/s and are separated by a random gap t1. Two attackers take part, h5 on access switch s12 and h6 on s13, and start 50 s into the run. Table 1 gives each attack's parameters.

Table 1. The seven attacks (per attacker; two attackers).

Attack	Source addresses	S (flows)	t1 (s)	M (flows)	Packets per refresh	Packet size	Other	Trained on by
G2	attacker's own	100	5.5–9.0	1,700	1	64 B	–	C2
Spoof	random, spoofed	100	5.5–9.0	1,700	1	64 B	–	C2
Pulse	attacker's own	300	5.0–8.0	1,700	1	64 B	60 s on, 30 s off; new flows each on period	C3
Mimic	pool of 256	100	6.0–9.5	1,700	1–3	benign	–	C1
Aware	pool of 256	100	2.0–3.0	1,700	1–3	benign	tuned against our earlier defenses	none
Slow	random, spoofed	45	8.0–9.8	1,700	1–2	benign	–	none
EVX	attacker's own	100	5.5–9.0	628	1	64 B	kept below FEVER's volume limit	none

S: new flows per burst. t1: gap between bursts, uniform in the range. M: live flows held per attacker. Pool of 256: addresses drawn from one /24 outside the network (10.254.x.0/24). Spoofed: random addresses from 1.0.0.0 to 222.255.255.255, a new one per flow. Benign packet size: drawn from the benign traffic's size distribution. G2 is DMS-BTT's second parameter group; Spoof and EVX use its S and t1. EVX's M is 0.8 times FEVER's lowest per-switch volume limit on T1B (784.5 flows on s12). Trained on by: the controller whose training data held the attack; none: no controller saw it.

2. Main observations

(1) **The preregistered tests mostly favour FEVER.** 25 of the 33 tests are supported (Table 2). The DMS-BTT correction of amendment 3 changed no test outcome.

(2) **FEVER harms almost no benign flows.** On T1B, FEVER harmed 0.011 % of benign flows on average, against 1.0 % to 5.2 % for the six baselines. H1 is supported against all six. In the benign-only runs FEVER harmed no flow and blocked no benign host (H7 passes).

(3) **FEVER keeps new connections working.** Mean new-connection success during attacks on T1B was 84.2 % with FEVER, against 21.0 % to 45.6 % for the baselines. H3 is supported against all six. Per attack, FEVER kept 97–99 % success on G2, Spoof, Pulse, Mimic and EVX.

(4) **FEVER holds attack occupancy lowest against five of six baselines.** On T1B the attacker held 6.8 % of the table under FEVER, against 24.2 % to 35.5 % under the baselines. FTODefender is the exception: it holds G2, Pulse and EVX lower than FEVER (0.3, 0.2 and 0.2 % against 4.8, 1.9 and 17.7 %), so H2 is not supported against it.

(5) **The largest margins are on the low-rate attacks Mimic and Aware.** Occupancy under FEVER was 4.9 and 14.1 %, against 33–49 % for the baselines. FEVER also beat every baseline retrained on the three controllers' data pooled (H4: 9.5 % against 20.6–39.1 %).

(6) **FEVER does not defend against Slow and EVX.** On these two attacks no H5 test on T1B is supported. Slow is defended by no method (about 20 % occupancy for all). EVX defeats FEVER (17.7 %), while FTODefender, DMS-BTT and FloRa hold it at 0.2–14.3 %. On UHt, the trace no controller trained on, FEVER beat FTMaster, DMS-BTT and rDefender on every pair (7.7 % against 37.0–40.6 %).

(7) **Federation helps the detector on attacks a controller never saw.** Offline, recall on attacks seen only by another controller rose from 75.8 % (C3 alone) to 90.5 %, and on Mimic from 0.2 % to 60.9 %. This equals pooled XGBoost trained on the raw rows (F1 86.5 against 86.8), at 2.6 GB of exchanged histograms and 38 s of training (Table 4, Fig. 3).

(8) **On the testbed, federation's benefit is clear on Mimic but not significant overall.** With the federated detector, Mimic attackers were blocked at 80 s instead of 144 s and occupancy fell from 15.7 to 4.9 %. Over Mimic and Aware, H6 is not supported (8 of 10 pairs, Holm $p = 0.38$).

(9) **Each part of FEVER contributes (Fig. 2).** Without ingress blocking, occupancy rises to 23–35 % on all three attacks tested. Without the volume rule, Aware rises from 14.1 to 42.6 %. Without the detector, blocks on G2 and Mimic come 35–55 s later. Without sharing, Mimic rises from 4.9 to 15.7 %.

(10) **The baselines are implemented faithfully.** Tested as their papers test (on the attacks the training controller saw), every reimplemented baseline reaches about 99.8 % accuracy, precision, recall and F1, at or above the values in its paper (Table 3). Over all T1B attacks, FEVER's F1 is 86.8 against 78.5 for each baseline trained on one controller's data.

(11) **FEVER is light on the controller.** Mean decision time was 20.8 ms, against 35–103 ms for the baselines, and controller CPU 62 % against 70–78 %. Its memory use was the highest (253 MB against 191–233 MB).

Table 2. Outcome of the preregistered tests: supported or not (paired wins of FEVER / pairs).

Hypothesis	Trace	FTMaster	FTOP	FTODefender	FloRa	DMS-BTT	rDefender
H1 Harm	T1B	yes (17/19)	yes (19/19)	yes (19/19)	yes (19/19)	yes (16/19)	yes (19/19)
H2 Attack occupancy	T1B	yes (19/19)	yes (19/19)	no (13/19)	yes (19/19)	yes (19/19)	yes (19/19)
H3 New-connection success	T1B	yes (19/19)	yes (19/19)	yes (18/19)	yes (19/19)	yes (19/19)	yes (19/19)
H4 Occupancy, pooled baselines	T1B	yes (10/10)	yes (10/10)	yes (10/10)	yes (10/10)	yes (10/10)	–
H5 Occupancy, Slow and EVX	T1B	no (5/10)	no (8/10)	no (2/10)	no (4/10)	no (3/10)	no (9/10)
H5 Occupancy, unseen trace	UHt	yes (9/9)	–	–	–	yes (9/9)	yes (9/9)

H3 is a non-inferiority test (FEVER's probe success above the comparator's minus 5 points). H4 compares FEVER with each baseline trained on pooled data, on Mimic and Aware. H5 on UHt was preregistered against FTMaster, DMS-BTT and rDefender only. H6 (FEVER against FEVER without sharing, Mimic and Aware): 8/10, not supported.

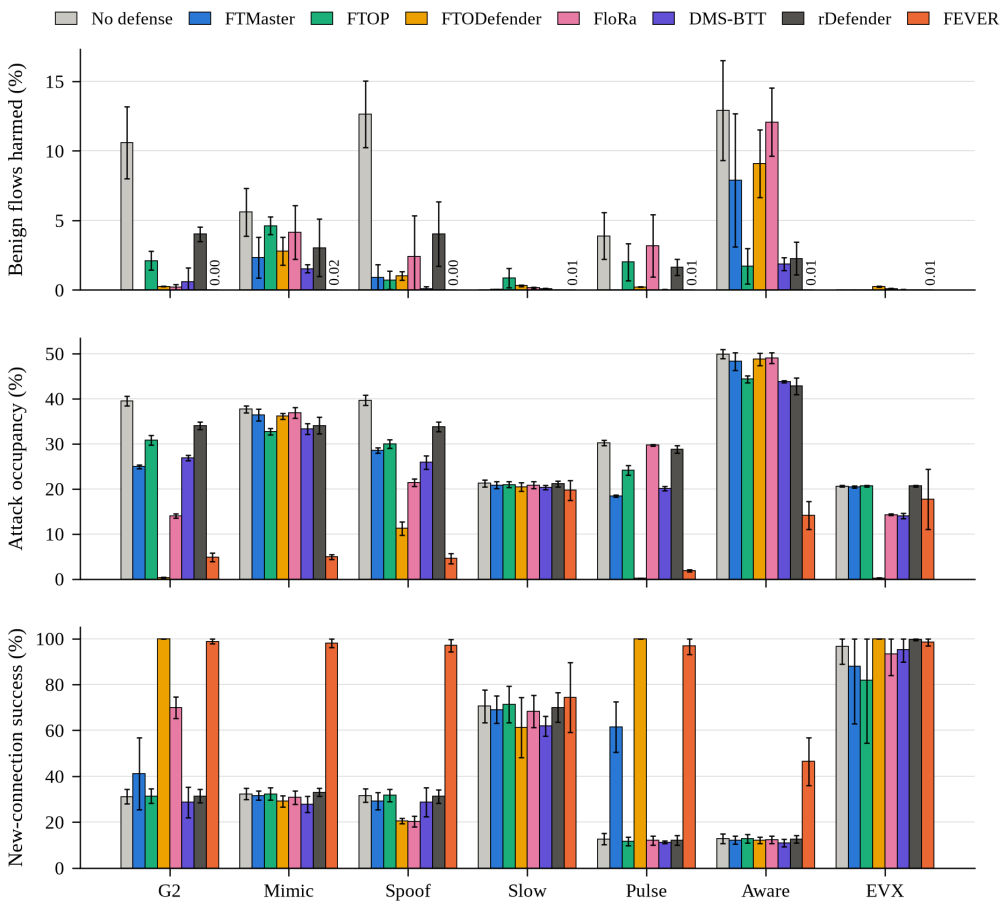


Fig. 1. Benign flows harmed, attack occupancy and new-connection success on T1B, per attack (mean and s.d. over seeds).

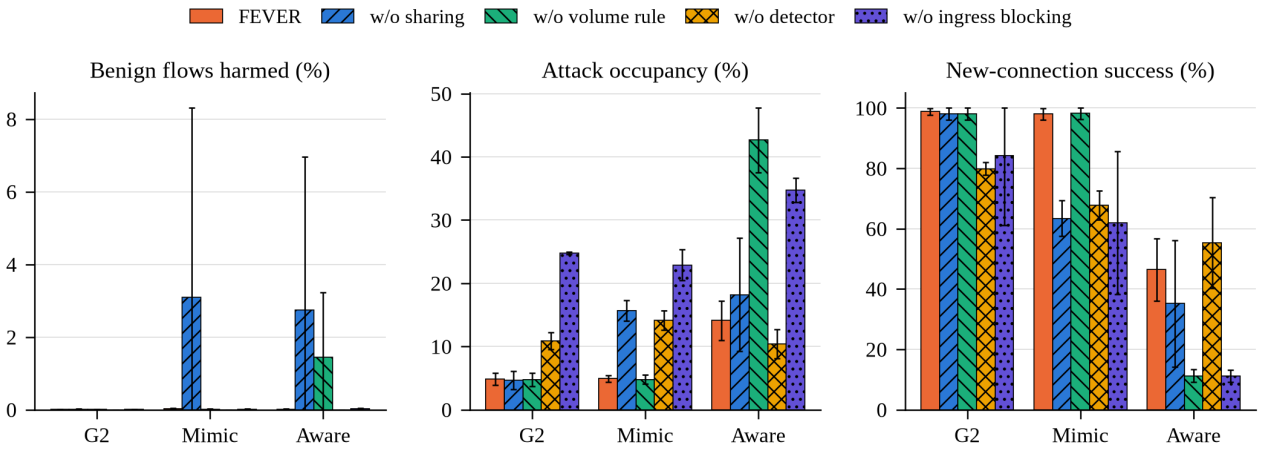


Fig. 2. FEVER against its four ablations on T1B.

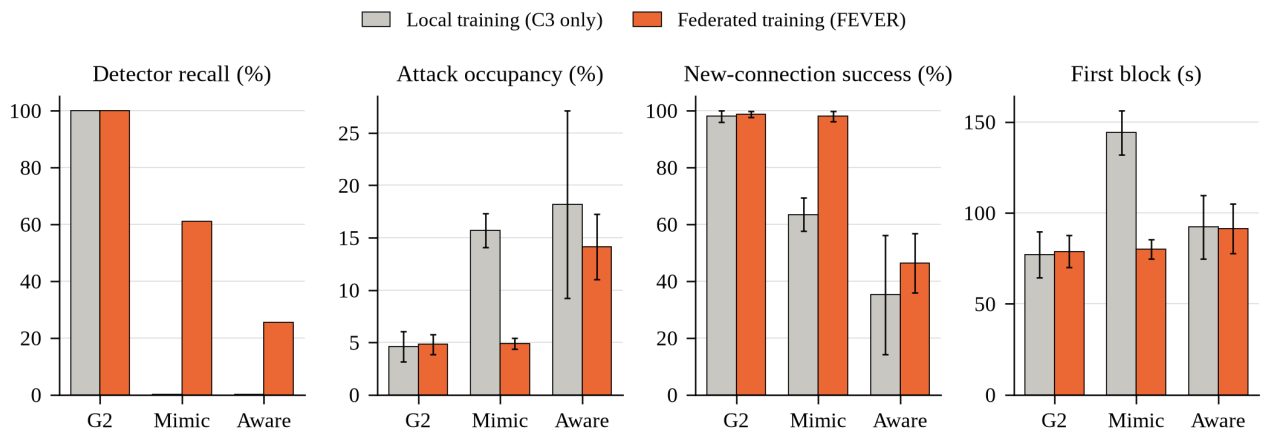


Fig. 3. Detector trained on C3's data only against the lossless federated detector: offline recall and the testbed outcome.

Table 3. Detection (%) of each classifier: values reported in each paper, and ours on T1B test rows.

Method	Paper: Acc. / Prec. / Rec. / F1	Ours, attacks seen in training	Ours, all T1B attacks
FTMaster	92.5 / – / 89.5 / 91.6	99.8 / 99.8 / 99.9 / 99.8	75.3 / 99.8 / 64.7 / 78.5
FTOP	98.9 / 99.2 / 99.0 / 99.1	99.8 / 99.8 / 99.9 / 99.8	75.4 / 99.8 / 64.7 / 78.5
FloRa	99.5 / – / 98.8 / 99.9	99.8 / 99.8 / 99.9 / 99.9	75.3 / 99.8 / 64.7 / 78.5
DMS-BTT	92.4 / 92.5 / 83.1 / 87.6	99.8 / 99.8 / 99.9 / 99.9	75.4 / 99.8 / 64.7 / 78.5
FEVER	–	99.8 / 99.7 / 99.9 / 99.8	83.7 / 99.7 / 76.8 / 86.8

Ours: baselines trained on C3's rows at their paper's threshold, FEVER federated. Attacks seen in training: G3 and Pulse. The papers' values come from their own datasets; FloRa's reported F1 does not agree with its own false-negative rate. FTODefender's scoring ran out of memory and rDefender has no classifier.

Table 4. Cost and accuracy of training the detector across the three controllers.

Training	Shared	Data sent (MB)	Time (s)	F1 (%)	Recall, other controllers (%)	Recall, Mimic (%)
C3 only	nothing	0	–	78.4	75.8	0.2
FedAvg	weights	0.4	32	74.7	70.7	0.1
FedProx	weights	0.4	27	75.9	71.4	1.1
Federated XGBoost bagging	trees	173	212	78.5	75.8	0.1
Lossless federated XGBoost (FEVER)	histograms	2,601	38	86.8	90.5	60.9
Pooled XGBoost	rows and labels	27.6	–	86.5	90.5	61.1

3. Where FEVER falls short

- Slow is undefended and EVX defeats FEVER; FTODefender holds G2, Pulse and EVX lower than FEVER.
- Federation's benefit on the testbed is not significant over Mimic and Aware (H6), only clear on Mimic.
- The detector's false alarms on T1B benign rows are 0.79 %, against the 0.1 % budget its threshold was set for. It failed its preregistered false-alarm rule and was deployed under an amendment.
- Pooling raw data gives the baselines a higher F1 than FEVER at their paper's threshold (90–92 against 86.8), at 2.6–3.7 % false alarms; at FEVER's false-alarm rule they reach 81–84.
- Ingress blocking contributes most of FEVER's occupancy reduction and is not new; the novelty lies in the federated detector and the evaluation.
- Scale is small: one machine, three controllers, 300 s runs, 3 to 5 seeds. The baselines are reimplementations with documented deviations (for example DMS-BTT's tuned variant and FTODefender's block rule), and three amendments were made to the preregistration.

4. Next steps

- Secure aggregation of the exchanged histograms and a threshold set from all three controllers' benign data (plan drafted: docs/secagg_plan.md).
- A time-to-protection experiment: a federation round during an attack, after which the controller replaces its local detector with the federated one.
- The manuscript for NetSoft 2027, framed on the federated detector and the preregistered evaluation, with the limitations above and the missing related work (MWD-CFM and Tang et al., 2026).